Holoscan Sensor Bridge Bring-up Notes

NVIDIA AGX Thor + Agilex 5 Modular Development Kit

This document records the bring-up steps, issues encountered, debug observations, and fixes used while running the Altera/NVIDIA Holoscan Sensor Bridge demo with NVIDIA AGX Thor as host and Agilex 5 Modular Development Kit as the FPGA platform.

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1. Hardware and Software Setup

Host

- NVIDIA Jetson AGX Thor Developer Kit
- JetPack 7.2 / L4T R39.2

Verify with:

```
cat /etc/nv_tegra_release
```

Expected:

```
# R39 (release), REVISION: 2.0
```

FPGA Boards / Designs Used

Two Agilex 5 designs were tested.

10GbE Design

```
AGX_5E_065B_Modular_DevKit_HSB_MIPI_10GbE.sof
```

Repository path:

```
fpga/altera/AGX_5E_065B_Modular_DevKit_HSB_MIPI_10GbE
```

25GbE Design

```
AGX_5E_065A_Modular_DevKit_HSB_MIPI_25GbE.sof
```

Repository path:

```
fpga/altera/AGX_5E_065A_Modular_DevKit_HSB_MIPI_25GbE
```

Repository Used

```
git clone -b altera-release-2.6.0 https://github.com/altera-fpga/holoscan-sensor-bridge.git
cd holoscan-sensor-bridge
git lfs pull
```

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2. Docker Build and Demo Container

On AGX Thor:

```
cd /home/alterademo/holoscandemo25G/holoscan-sensor-bridge
sh docker/build.sh --igpu
```

The first Docker build can take a significant amount of time.

After the image is built:

```
xhost +
sh docker/demo.sh
```

Inside the container, tools such as hololink-enumerate are installed in:

```
/usr/local/bin/hololink-enumerate
```

Use:

```
hololink-enumerate
```

Do not use this path inside the demo container:

```
./tools/enumerate/hololink-enumerate
```

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3. 10GbE Bring-up

3.1 Program FPGA

Program the Group B / 10GbE SOF:

```
AGX_5E_065B_Modular_DevKit_HSB_MIPI_10GbE.sof
```

Example Quartus command:

```
quartus_pgm -c 1 -m jtag -o "p;AGX_5E_065B_Modular_DevKit_HSB_MIPI_10GbE.sof"
```

3.2 Check Thor Ethernet Link

On AGX Thor host:

```
for i in 0 1 2 3; do
  echo "---- mgbe${i}_0 ----"
  sudo ethtool mgbe${i}_0 | grep -E "Speed|Link detected"
done
```

Expected for a working 10G setup:

```
---- mgbe0_0 ----
Speed: 10000Mb/s
Link detected: yes
```

3.3 Configure Thor IP

```
EN0=mgbe0_0

sudo nmcli connection delete hololink-$EN0 2>/dev/null || true
sudo nmcli con add con-name hololink-$EN0 ifname $EN0 type ethernet ip4 192.168.0.101/24
sudo nmcli connection modify hololink-$EN0 +ipv4.routes 192.168.0.2/32
sudo nmcli connection up hololink-$EN0
```

Check route:

```
ip route | grep 192.168.0
```

Expected:

```
192.168.0.0/24 dev mgbe0_0 ...
192.168.0.2 dev mgbe0_0 ...
```

Ping FPGA:

```
ping 192.168.0.2
```

Expected:

```
64 bytes from 192.168.0.2
```

3.4 Enumerate HSB

Inside Docker:

```
hololink-enumerate
```

Successful output example:

```
mac_id=CA:FE:C0:FF:EE:10
hsb_ip_version=0x2603
fpga_crc=0x0
ip_address=192.168.0.2
fpga_uuid=7b1fa8c7-31aa-44b6-abcc-eac134461fdc
serial_number=01000000000000
interface=mgbe0_0
board=N/A
```

This confirms:

```
Ethernet link: OK
HSB enumeration: OK
FPGA reachable: OK
```

3.5 Run Camera Demo

Single camera:

```
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30
```

Camera 1:

```
python3 examples/linux_agx5_player.py --cam 1 --lines 1080 --frame-rate 30
```

Stereo:

```
python3 examples/linux_agx5_player_stereo.py
```

Headless test:

```
python3 examples/linux_agx5_player.py --headless --frame-limit 100 --cam 0 --lines 1080 --frame-rate 30
```

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4. Issues Encountered During 10G Bring-up

Issue 1: No Ethernet Link / No Carrier

Observed:

```
Speed: 10000Mb/s
Link detected: no
```

or:

```
NO-CARRIER
Destination Host Unreachable
```

Ping output:

```
From 192.168.0.101 icmp_seq=1 Destination Host Unreachable
```

Meaning:

- 192.168.0.101 is Thor’s own IP.

- Thor is saying it cannot reach the FPGA at 192.168.0.2.
- This is a physical/link issue, not Docker or Holoscan.

Resolution:

- Confirm correct FPGA SOF.
- Confirm correct SFP/QSFP cable.
- Confirm correct breakout leg.
- Confirm correct Thor interface.
- Use the 10G FPGA design when Thor is in default 10G mode.

Issue 2: Wrong Tool Path

Command failed:

```
./tools/enumerate/hololink-enumerate
```

Error:

```
No such file or directory
```

Resolution:

```
which hololink-enumerate
```

Output:

```
/usr/local/bin/hololink-enumerate
```

Use:

```
hololink-enumerate
```

Issue 3: Camera I2C Transaction Error

Observed:

```
hololink._hololink.TransactionError: i2c_transaction i2c_address=0x37
```

and for camera 1:

```
hololink._hololink.TransactionError: i2c_transaction i2c_address=0x1a
```

Meaning:

- FPGA/HSB path is working.
- Camera sensor is not responding over I2C.
- On early production Agilex 5 MDK boards, this can be caused by the carrier MAX10 firmware not applying power to the MIPI connectors.
- Without MIPI connector power, camera I2C communication is impossible and Holoscan applications report transaction errors.

Checks performed:

- Camera cable orientation.
- Pin 1 to pin 1 alignment.
- Correct MIPI connector.
- Camera power/reset sequencing.
- Reprogram FPGA after camera connection.

Root cause found during this bring-up:

- Early production MDK boards, both Group A and Group B, shipped with a bug in the MAX10 image.
- The buggy MAX10 image does not apply power to the MIPI connectors.
- Customer release production boards are expected to ship with the fixed MAX10 image and should be plug-and-play.
- Currently available / early boards may need the MAX10 image reflashed.

Resolution that fixed the camera I2C transaction errors:

1. Switch the MAX10 onto the JTAG chain by setting SW4 on the carrier board to ON. This is the single switch nearest the HDMI connector.
2. Use Quartus Programmer GUI to reflash MAX10 with:

```
max10_top_rtl_v1p1p6_fw_v2p0p1.pof
```

3. Power off the board.
4. To see the FPGA on the JTAG chain again, set SW4 back to OFF.

After reflashing the MAX10 image and restoring the normal JTAG chain, camera I2C configuration succeeded.

Additional note:

- A problem with initial production MDK batches is that the USB hub controlling the carrier micro USB is susceptible to ESD.
- This is expected to be resolved in future batches.
- If the carrier micro USB is not detected, for example Windows fails to read descriptors, use the standard JTAG connector and a Byte Blaster instead.

Successful camera configuration log:

```
Configuring camera with frame format: Width=1920, Height=1080, Framerate=30, Pixel
```

```
Format=PixelFormat.RAW_10
Starting camera
Stopping camera
```

Issue 4: GUI / Holoviz Failure

Observed:

```
Failed to initialize glfw
Authorization required, but no authorization protocol specified
XDG_RUNTIME_DIR is invalid or not set in the environment
```

Meaning:

- Camera and streaming path may be working.
- Display permission is failing.

Resolution on Thor host:

```
xhost +local:root
xhost +local:docker
xhost +
```

Restart Docker:

```
sh docker/demo.sh
```

Inside Docker:

```
export XDG_RUNTIME_DIR=/tmp/runtime-root
mkdir -p $XDG_RUNTIME_DIR
chmod 700 $XDG_RUNTIME_DIR
```

Then run the demo again.

Issue 5: Kernel Receiver Buffer Warning

Observed:

```
Kernel receiver buffer size is too small; performance will be unreliable.
Resolve this with "echo 2621440 | sudo tee /proc/sys/net/core/rmem_max"
```

Resolution on Thor host:

```
echo 2621440 | sudo tee /proc/sys/net/core/rmem_max
```

For 4K mode, use a larger buffer:

```
echo 10420224 | sudo tee /proc/sys/net/core/rmem_max
```

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5. 25GbE Bring-up

5.1 Initial 25G Problem

Initially the 25G design did not link.

Observed:

```
Speed: 10000Mb/s
Link detected: no
```

Root cause:

- AGX Thor defaults to 10GbE x4 mode on the QSFP/MGBE ports.
- The 25G FPGA design requires 25GbE x4.
- Thor does not support runtime switching between 10G and 25G using normal Linux network settings.
- nmcli, IP settings, Docker, and FPGA reprogramming alone cannot switch Thor to 25G.

5.2 25G Switching Method

Altera provided capsule files for switching Thor bootloader/QSPI configuration between 10G and 25G.

Required files:

```
Tegra_Thor_BL_7.2_10G.Cap
Tegra_Thor_BL_7.2_25G.Cap
```

JetPack requirement:

```
JetPack 7.2 / L4T R39.2
```

Verified using:

```
cat /etc/nv_tegra_release
```

Output:

```
# R39 (release), REVISION: 2.0
```

5.3 Current Slot Before Switching

Before switching, check:

```
sudo nvbootctrl get-current-slot
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

Output before switching:

```
1
10000
10000
10000
10000
```

Recorded:

```
slot 1 = 10G
```

5.4 Apply 25G Capsule

Copy capsule files to Thor home directory:

```
cd ~
ls -lh *.Cap
```

Then apply the 25G capsule:

```
sudo nv_bootloader_capsule_updater.sh -q ./Tegra_Thor_BL_7.2_25G.Cap
sudo reboot
```

After reboot:

```
sudo nvbootctrl get-current-slot
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

Output:

```
0
25000
25000
25000
25000
```

Recorded:

```
slot 0 = 25G
slot 1 = 10G
```

5.5 Reverting Back to 10G

Since 10G is slot 1:

```
sudo nvbootctrl set-active-boot-slot 1
sudo reboot
```

Verify:

```
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

Expected:

```
10000
10000
10000
10000
```

5.6 Switching Back to 25G

Since 25G is slot 0:

```
sudo nvbootctrl set-active-boot-slot 0
sudo reboot
```

Verify:

```
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

Expected:

```
25000
25000
25000
25000
```

??

6. 25G FPGA Demo Flow

Once Thor is in 25G mode:

```
slot 0 active
MGBE speeds = 25000
```

Program the 25G FPGA:

```
AGX_5E_065A_Modular_DevKit_HSB_MIPI_25GbE.sof
```

Check link:

```
for i in 0 1 2 3; do
  echo "---- mgbe${i}_0 ----"
  sudo ethtool mgbe${i}_0 | grep -E "Speed|Link detected"
done
```

Expected:

```
Speed: 25000Mb/s
Link detected: yes
```

Configure IP:

```
EN0=mgbe0_0

sudo nmcli connection delete hololink-$EN0 2>/dev/null || true
sudo nmcli con add con-name hololink-$EN0 ifname $EN0 type ethernet ip4 192.168.0.101/24
sudo nmcli connection modify hololink-$EN0 +ipv4.routes 192.168.0.2/32
sudo nmcli connection up hololink-$EN0
```

Ping:

```
ping 192.168.0.2
```

Enter container:

```
cd /home/alterademo/holoscandemo25G/holoscan-sensor-bridge
xhost +
sh docker/demo.sh
```

Inside container:

```
hololink-enumerate
```

Run demo:

```
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30
```

Stereo:

```
python3 examples/linux_agx5_player_stereo.py
```

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7. Current Known Good State

10G

```
slot 1 = 10G
FPGA image = AGX_5E_065B_Modular_DevKit_HSB_MIPI_10GbE.sof
MGBE speed = 10000
HSB enumerate = PASS
Camera I2C = PASS
Headless streaming = PASS
GUI requires xhost/XDG setup
```

25G

```
slot 0 = 25G
FPGA image = AGX_5E_065A_Modular_DevKit_HSB_MIPI_25GbE.sof
```

```
MGBE speed = 25000
25G mode enabled successfully
```

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8. Useful Debug Commands

Check JetPack / L4T

```
cat /etc/nv_tegra_release
```

Check MGBE Speeds

```
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

Check Active Boot Slot

```
sudo nvbootctrl get-current-slot
```

Check Link Status

```
for i in 0 1 2 3; do
  echo "---- mgbe${i}_0 ----"
  sudo ethtool mgbe${i}_0 | grep -E "Speed|Link detected"
done
```

Configure IP

```
EN0=mgbe0_0

sudo nmcli connection delete hololink-$EN0 2>/dev/null || true
sudo nmcli con add con-name hololink-$EN0 ifname $EN0 type ethernet ip4 192.168.0.101/24
sudo nmcli connection modify hololink-$EN0 +ipv4.routes 192.168.0.2/32
sudo nmcli connection up hololink-$EN0
```

Ping FPGA

```
ping 192.168.0.2
```

Enumerate HSB

```
hololink-enumerate
```

Run Single Camera

```
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30
```

Run Headless

```
python3 examples/linux_agx5_player.py --headless --frame-limit 100 --cam 0 --lines 1080 --frame-rate 30
```

Run Stereo

```
python3 examples/linux_agx5_player_stereo.py
```

Fix GUI Permission

On Thor host:

```
xhost +local:root
xhost +local:docker
xhost +
```

Inside Docker:

```
export XDG_RUNTIME_DIR=/tmp/runtime-root
mkdir -p $XDG_RUNTIME_DIR
chmod 700 $XDG_RUNTIME_DIR
```

Fix Receiver Buffer Warning

```
echo 2621440 | sudo tee /proc/sys/net/core/rmem_max
```

For larger frames:

```
echo 10420224 | sudo tee /proc/sys/net/core/rmem_max
```

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9. Lessons Learned

1. Thor default QSFP mode is 10G.
2. The 25G FPGA image will not link unless Thor is booted in 25G mode.
3. 10G/25G switching is not runtime-configurable using nmcli.
4. Capsule-based slot switching allows 10G/25G switching without wiping the Linux rootfs.
5. hololink-enumerate proves FPGA/HSB communication.
6. Camera I2C errors are separate from Ethernet link errors.
7. GUI/Holoviz failures are often X11 authorization problems, not camera/FPGA problems.
8. Always use Ctrl+C to stop apps, not Ctrl+Z.
9. .sof programming is volatile; reprogram FPGA after a board power cycle.
10. Record boot slot mapping before switching modes.

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10. Final Slot Mapping From This Bring-up

```
slot 0 = 25G
slot 1 = 10G
```

Switch to 25G:

```
sudo nvbootctrl set-active-boot-slot 0
sudo reboot
```

Switch to 10G:

```
sudo nvbootctrl set-active-boot-slot 1
sudo reboot
```